Your professional
photo here
(replace this)

Abdelhak Bourdim

Senior Embedded Software Engineer — C/C++ • RTOS • Embedded Linux

Date of birth: 20 August 1969 **Nationality:** French **Status:** Married

Permit: Driving license B • Eligible for Swiss G permit (cross-border) **Email:** abdelhak.bourdim@gmail.com

Phone: +33 6 19 51 51 73 **Location:** Paris area, France

LinkedIn: linkedin.com/in/abdelhak-bourdim-96416122 **Website:** workshop-diy.org

Mobility: Western Switzerland (Geneva / Lausanne) — Available immediately

PROFILE

Embedded software specialist with **over 20 years of experience** developing firmware, middleware, and application layers — bare metal, RTOS, and Embedded Linux — for critical and industrial systems. Designing low-level drivers, communication stacks (CAN, BLE, USB, TCP/IP), and embedded applications on ARM Cortex, STM32, Microchip, and NXP architectures.

Delivered certified solutions for **aerospace** (Safran, Thales — DO-178), **medical devices** (GE Healthcare, Sorin — IEC 62304), **automotive** (Valeo, Arkamys), **IoT cybersecurity** (Enedis — Linky), **payment systems** (Cartadis), and **V2G smart charging** (E2-CAD — ISO 15118). MISRA C compliant development.

PROFESSIONAL EXPERIENCE

E2-CAD — Eragny, France

May 2025 – Mar. 2026

Embedded Software Engineer — EV Charging / V2G

EV Charging Station — Iotecha IP

- Reverse engineering and source code analysis (static and dynamic architecture)
- Embedded code review on 3 boards (Microchip PIC, STM32), debug tools setup (ST-Link, nsprog)
- Communication protocol analysis between Linux modules and embedded boards (RS232, I2C, Saleae)
- Python extension development and real-time HTML web report generation
- Test bench design and setup: Chargebyte tools integration, EV simulator
- Charging station debug and commissioning (PLC, HomePlug Green PHY, ISO 15118, OCPP)

Env: Embedded Linux, C/C++, Microchip PIC, STM32, PLC, HomePlug Green PHY, ISO 15118, OCPP, CAN, TCP/IP, Saleae, HackRF One, Python, Bash

Safran — Creteil, France

Mar 2024 – May 2025

Embedded Software Engineer — Aerospace (SSPC)

Solid State Power Controller — Low-level driver updates

- Development and update of low-level C drivers for the SSPC system
- Implementation of logging tools for diagnostics and exchange traceability
- Writing and execution of on-target validation tests with Lauterbach TRACE32
- Bash/Awk scripting for test process automation

Env: C, Lauterbach TRACE32, Bash, Awk

Asterios Technologies — Massy, France

Aug 2023 – Feb 2024

Embedded Software Engineer — Aerospace (SOM Testing)

- C development of validation tests for the SOM module (T1042 processor)
- Processor errata analysis and software workaround identification
- Test documentation writing compliant with project requirements

Env: C, Bash, Python, Lauterbach TRACE32, Polarion

Thales — Gennevilliers, France

May 2022 – Jul 2023

Embedded Software Engineer — Defense Radio

LPM + N-MMR Projects — Mission preparation & Multi Mode Receiver

- Application layer development under Embedded Linux in C/C++ for 2 projects
- Configuration management and automation scripts (Bash, Curl, Tshark)
- Development and execution of HLT (High Level Tests)
- Bug fixing and network protocol analysis (Websockets)

Env: Embedded Linux, C/C++, Bash, Websockets, Curl, Tshark, Python

Valeo — Creteil, France / Egypt

Mar – Apr 2022

Embedded Software Engineer — Test Audit

- Audit of test processes and tools (NI Mastre, Vector CANoe)
- Automation proposals and client/system requirements review

Env: NI Mastre, Vector CANoe

Arkamys — Levallois-Perret, France
Jul 2021 – Feb 2022

Embedded Software Engineer — Automotive Audio

- GStreamer plugin and CAN driver (ELM327) development, diagnostic tools
- Unit and integration testing (Valgrind), build system Buildroot

Env: Embedded Linux, C/C++, Buildroot, GStreamer, Valgrind, CAN, Python, Jira

PROFESSIONAL EXPERIENCE (CONTINUED)

Thales — Velizy-Villacoublay, France
Nov 2020 – Jun 2021

Embedded Software Engineer — Autonomous Train (ATO)

- Application migration from 32-bit to 64-bit on Embedded Linux platform
- IPC communication protocol definition and implementation
- Network diagnostic tools development and integration testing

Env: Embedded Linux, C, UDP, TCP, Wireshark, Tshark, Python, Jira

PK Vitality — Paris, France
Jun – Sep 2020

Embedded Software Engineer — Medical Device (CGM)

- HAL development (AFE AD5940, PMIC DA9070), QSPI driver, BLE NRF52840
- STM32 TouchGFX touchscreen, FreeRTOS, low power mode
- In vitro testing and Doxygen documentation (IEC 62304 compliant)

Env: STM32L4S9, NRF52840, C, C++, FreeRTOS, BLE, TouchGFX, Doxygen

Zodiac Aerospace (Safran) — Montreuil / Vancouver, Canada
Feb 2017 – Mar 2020

Embedded Software Engineer — Aerospace (SSPC A350)

- Low-level layer adaptation: CAN, RS485 drivers, thermal memory (dsPIC33)
- uAFDX to CAN gateway implementation and diagnostic functions
- SSPC A350 application layer development and optimization
- Validation testing and aerospace standards-compliant documentation
- Flight test campaign at UBC Vancouver, Canada (DO-178 certification)

Env: Microchip dsPIC33, MPLAB, TI Concerto, CCS Studio, C

Enedis — Nanterre, France
May 2015 – Sep 2016

Embedded Cybersecurity Engineer — Smart Grid

- Linky concentrator security diagnostics and audit
- Penetration testing on Embedded Linux systems (Kali Linux, Nmap)
- Protocol sniffing and fuzzing (Wireshark, Tshark, Scapy, SDR)
- Code audit and review with manufacturers (Sonar, PC Lint)

Env: Kali Linux, Embedded Linux, Nmap, Wireshark, Scapy, SDR, Python, Bash

Sorin — Clamart, France
Nov 2014 – Apr 2015

Embedded Software Engineer — Medical (BLE Pacemaker)

- Bluetooth Smart module validation, Serial Port Service Profile development
- Smartphone communication protocol development and Android application

Env: Cortex M0, Keil, C, BLE, Android

General Electric Healthcare — Buc / USA
2012 – 2014

Embedded Software Engineer — Medical Imaging (X-Ray)

- CANOpen stack integration (slave/master), CAN driver and service development
- CMX-RTX kernel porting, USA team technical support and on-site visits

Env: TI TMS320, ARM Cortex A8, VxWorks, C, C++

Photomaton / BCM / Vertical M2M / PGS — Paris
2010 – 2012

Payment systems and remote metering — Full embedded + PC development on NXP LPC (USB, RFID, GSM/GPRS, Bluetooth)

Cartadis — Fontenay-sous-Bois, France
Oct 2001 – Dec 2009

8 years — Access control and payment terminals: drivers, embedded applications, USB/TCP/IP stacks, DES/RC4 encryption (NEC V853, Hitachi H8S, NXP LPC, ATMEL AT91)

TECHNICAL SKILLS

Languages	C, C++, Python, Bash, Assembly, Perl, Awk
OS / RTOS	Embedded Linux, VxWorks, FreeRTOS, UCOSII, Tnkernel
Microcontrollers	ARM Cortex (M0, M4, A8), STM32, NRF52, ESP32, Microchip PIC/dsPIC, NXP LPC, TI DSP/Concerto
Protocols / Bus	CAN, CANOpen, SPI, I2C, RS232, RS485, UART, USB, BLE, AFDX, Ethernet, TCP/UDP

Stacks	TCP/IP, USB Host/Device/HID, BLE, CANOpen, RFID, GStreamer
Security / Payment	Smart Card, RFID Mifare, DES, RC4, USB CCID, embedded pentesting
Tools	Lauterbach TRACE32, IAR, Keil, MPLAB, CCS Studio, Green Hills, Buildroot, CubeMx, TouchGFX
Management	Git, Synergy, Jira, Polarion, Doxygen, Wireshark, Saleae

STANDARDS & METHODS

DO-178 · MISRA C · IEC 62304 · ISO 15118 (V2G) · ISO 14443 (Smart Card) · ISO 15693 (RFID) · OCPP · V-Model · Agile / Scrum

EDUCATION

AFPA, Champs-sur-Marne, France
Engineer — Electronic and Computer Systems Design

Tenon Hospital, Paris, France
Postgraduate Diploma — Biomedical Instrumentation

University of Boumerdes, Algeria
Engineer — Electronics, Communication

LANGUAGES

French	*Native / Bilingual*
English	*Professional working*
Arabic	*Native / Bilingual*

INDUSTRY SECTORS

Aerospace (Safran, Thales, Zodiac)
Medical (GE Healthcare, Sorin, PK Vitality)
Automotive (Valeo, Arkamys)
Energy / Smart Grid (Enedis — Linky)
EV Charging / V2G (E2-CAD)
IoT / M2M (Vertical M2M)
Payment (Cartadis, Photomaton, BCM)

VOLUNTEER WORK

Workshop-DIY
Founder — Chelles, France — since 2020

Robotics & STEM workshops for youth (ages 6-12+)
ESP32 robot kit design (WiFi, BLE, servos, sensors)
18+ workshops: Arduino, micro:bit, AI, 3D printing
workshop-diy.org

References available upon request